Effects of IEP Training on Student Outcomes

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Purposes

The purpose of this study was to explore the impact of the IEP (Individualized Education Program)-Quality Tutorial, which was designed to improve the quality of IEPs, on student outcomes, including performance on state Reading and Math tests.

Method & Data Resources

Data Source

1. 80 Teachers in 12 districts participated in the IEP online training tutorial in Fall 2013.
2. 72 teacher remained in study by the end of 2014 (intervention: 34; control: 38).
3. Students who were taught by these 72 teachers and had scores from both years were included in the analyses, which were conducted for each content area.

Table 1. Numbers of Students with Reading and Math ISAT Scores for Intervention and Control Teachers

	Reading			Math		
	2013	2014	Both year	2013	2014	Both year
Intervention	187	171	164	186	167	159
High Use	49	47	44	48	47	43
Low Use	138	124	120	138	120	126
Control	148	133	124	148	133	124

Measures

State Reading and Math scaled scores and proficiency levels from the Illinois Standards Achievement Test (ISAT) in 2013 and 2014 were collected.

Procedures

1. Teachers were randomly assigned by district to either a control or an intervention group.
2. Teachers in the intervention group could access the online Tutorial training. Teachers in the control group obtained access to the tutorial only after the study was done.
3. Teachers provided the state test scores of their students in 2013 (school year 2012-13), and 2014 (school year 2013-14). These represented times before and after the IEPQ Tutorial training intervention.

Analysis

Descriptive analyses and ANCOVA were conducted by content area (reading and math).

Results

Reading

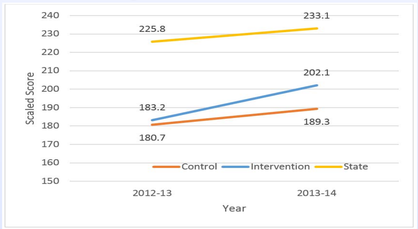


Figure 1. The mean of ISAT Reading Scaled Score in Each Year by Intervention and Control groups

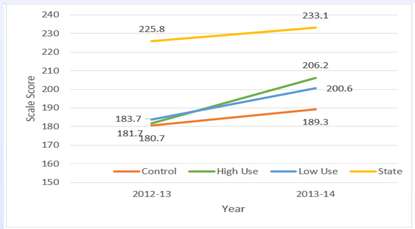
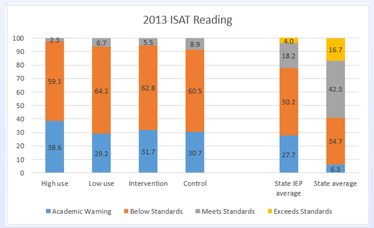


Figure 2. The mean of ISAT Reading Scaled Score in Each Year by High Use, Low Use, and Control groups



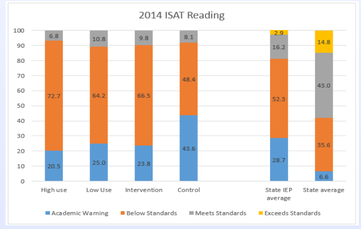


Figure 5. Percentage of Students at each Proficiency Level on ISAT Reading

Table 2. Summary of Analysis of Covariance on ISAT Reading by 2 Groups

Source	*SS*	*df*	*MS*	*F*	η_p^2
Group	7981.28	1	7981.28	74.92***	0.208
Covariate (2013 score)	121416.25	1	121416.25	1139.66***	0.800
Error	30363.15	285	106.54		
Total	163398.44	287			

Note. *** $p < .001$

Table 3. Summary of Analysis of Covariance on ISAT Reading by 3 Groups

Source	*SS*	*df*	*MS*	*F*	η_p^2
Group	9740.42	2	4870.21	48.35***	0.254
Covariate (2013 score)	122167.93	1	122167.93	1212.97***	0.810
Error	28604.02	284	100.72		
Total	163398.44	287			

Note. *** $p < .001$

Table 6 Chi-square Results for ISAT Reading

Variable	group	χ^2	p-value
Year[1]	2013	1.25	.54
	2014	**12.7**	**.002**
PF level	Academic Warning	**4.59**	**.0321**
	Below Standards	1.604	.2053
	Meet Standards	1.2462	.2643
	Exceeds Standards	0	1.0

[1]To test whether there is a significant association between the groups (intervention vs. control) and the proficiency level each year.

Math

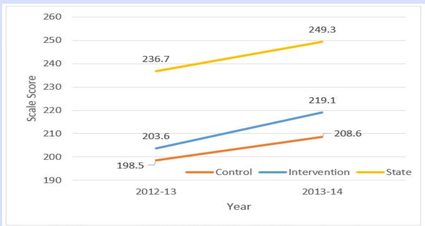


Figure 3. The mean of ISAT Math Scaled Score in each year by Intervention and Control groups

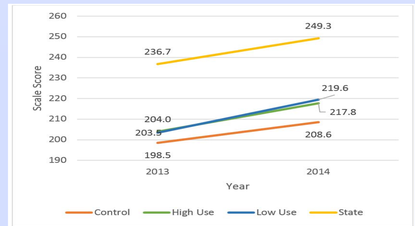
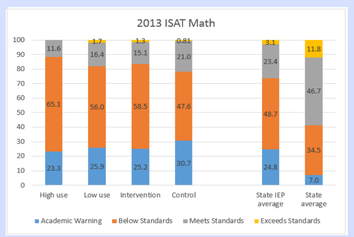


Figure 4. The mean of ISAT Math Scaled Score in each year by High Use, Low Use, and Control groups



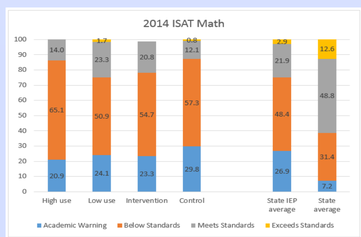


Figure 6. Percentage of Students at each Proficiency Level on ISAT Math

Table 4. Summary of Analysis of Covariance on ISAT Math by 2 Groups

Source	*SS*	*df*	*MS*	*F*	η_p^2
Group	2727.82	1	2727.82	16.14***	0.05
Covariate (2013 score)	134549.43	1	134549.43	796.03***	0.74
Error	47327.11	280	169.03		
Total	189594.50	282			

Note. *** $p < .001$

Table 5. Summary of Analysis of Covariance on ISAT Math by 3 Groups

Source	*SS*	*df*	*MS*	*F*	η_p^2
Group	2883.77	2	1441.89	8.53***	0.06
Covariate (2013 score)	134607.14	1	134607.14	796.15***	0.74
Error	47171.16	279	169.07		
Total	189594.49	282			

Note. *** $p < .001$

Table 7 Chi-square Results for ISAT Math

Variable	group	χ^2	p-value
Year[1]	2013	3.799	.284
	2014	4.443	.217
PF level	Academic Warning	.025	.874
	Below Standards	1.192	.275
	Meet Standards	4.333	.037
	Exceeds Standards	.000	1.000

[1]To test whether there is a significant association between the groups (intervention vs. control) and the proficiency level each year.

Conclusion

Scaled Score

This study found that students taught by the intervention teachers who received the IEPQ tutorial online training had higher scaled scores on both **Reading** and **Math** state assessments than students who were taught by teachers in the control group.

Proficiency Level

For **Reading**, in Year 2014, group was significantly related to proficiency level; this was not found in Year 2013.

The post-hoc test showed

1. the percentage of students in the control group in the proficiency level 1, Academic Warning, was significantly higher than the percentage in the Intervention group ($\chi^2 = 12.62$, $p = .0004$)
2. the percentage of students in the intervention group in proficiency level 2, Below Standards, was significantly higher than the percentage in the control group ($\chi^2 = 9.52$, $p = .0020$)
3. no significant result was found for proficiency level 3 (Meet Standards, $\chi^2 = 0.25$, $p = .6199$).

No relation was found between PF and Year on **Math**.